The Islamic (Hijri) Calendar and Time-Keeping in the Islamic World

What you will be able to do

1. Describe the Hijri calendar: its months, its year length and its starting point.
2. Explain why each month begins with the **sighting of the crescent Moon (hilal)**, and why Eid can fall on different days in different places.
3. Calculate how Ramadan shifts by about 11 days every year and passes through every season in about 33 years.
4. Convert roughly between Hijri (AH) and Gregorian (CE) years.
5. Appreciate the contributions of Islamic-era astronomers, whose star names we still use every night.

1. A purely lunar calendar

The Islamic calendar (*al-taqwim al-hijri*) is **purely lunar**:

- **12 months**, each beginning with the first visible crescent after New Moon;
- each month has **29 or 30 days**;
- one year = **354 or 355 days**, about **11 days shorter** than the solar year;
- **no extra (leap) month** is ever added — the Qur'an fixes the number of months at twelve and the older Arabian practice of inserting months (*nasi'*) was abandoned.

As a result, Islamic months are **not tied to the seasons**: Ramadan can come in winter, in the monsoon or in the heat of summer. Compare this with the Hindu calendar (Lecture 1.4), which adds an Adhik Maas precisely to avoid such drift.

The starting point: the Hijra (622 CE)

Year 1 begins with the **Hijra**, the migration of the Prophet Muhammad from Mecca to Medina in **622 CE**. Years are written **AH** (*Anno Hegirae*). The calendar was formally set up about 17 years later under the second Caliph, Umar ibn al-Khattab; 1 Muharram 1 AH corresponds to 16 July 622 (Julian calendar).

This year

1448 AH began on about 16–17 June 2026; 1449 AH is expected to begin around 6 June 2027.

2. The twelve months

#	Month	Meaning / notes
1	**Muharram**	"sacred"; Islamic New Year; **Ashura** on the 10th
2	Safar	

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(*continued*)

#	Month	Meaning / notes
3	**Rabi' al-Awwal**	"first spring"; birth of the Prophet (Milad-un-Nabi) observed on the 12th
4	Rabi' al-Thani	"second spring"
5	Jumada al-Ula	
6	Jumada al-Akhirah	
7	**Rajab**	sacred month
8	Sha'ban	Shab-e-Barat on the 15th night (observed in South Asia)
9	**Ramadan**	**month of fasting**, dawn to sunset
10	**Shawwal**	**Eid al-Fitr** on the 1st
11	Dhu al-Qa'dah	sacred month
12	**Dhu al-Hijjah**	month of the Hajj; **Eid al-Adha** (Bakrid) on the 10th

The four sacred months are Muharram, Rajab, Dhu al-Qa'dah and Dhu al-Hijjah. Some names describe seasons ("spring"; *Ramadan* from *ramad*, "scorching heat"). They were named when the old Arabian calendar was still kept roughly in step with the seasons.

3. The day begins at sunset; the week counts days

An Islamic day begins at **sunset**, not at midnight — the night of Eid comes *before* the day of Eid. Arabic weekday names mostly **count** the days instead of naming planets:

Day	Arabic	Meaning	Day	Arabic	Meaning
Sunday	al-Ahad	the first	Thursday	al-Khamis	the fifth
Monday	al-Ithnayn	the second	Friday	**al-Jumu'ah**	the day of **gathering**
Tuesday	ath-Thulatha	the third	Saturday	as-Sabt	the Sabbath (rest)
Wednesday	al-Arba'a	the fourth			

Compare with the planetary names in Lecture 1.4: both systems share the same 7-day week but name the days differently.

4. Moon sighting (Ru'yat al-Hilal)

A new Islamic month begins when the thin **crescent** (*hilal*) is first sighted in the western sky just after sunset, a day or two after the astronomical New Moon (Amavasya). The young Moon is very thin, faint and low, and sets soon after the Sun. It is usually visible only if

- the Moon is at least about **15–20 hours old** (time since New Moon), and
- it is at least **7°–10°** away from the Sun (the minimum is called the **Danjon limit**), and
- it sets at least ~40 minutes after the Sun, so the sky is dark enough while it is still up.

Clouds, dust and haze often hide it. That is why the start of Ramadan and the date of Eid are often announced only the evening before, and why they can differ by a day between, say, Saudi Arabia and India. In India, local **Ruet-e-Hilal** (moon-sighting) committees make the announcement. Some countries use astronomical calculation instead; Saudi Arabia's civil **Umm al-Qura** calendar is computed in advance.

The tabular Islamic calendar

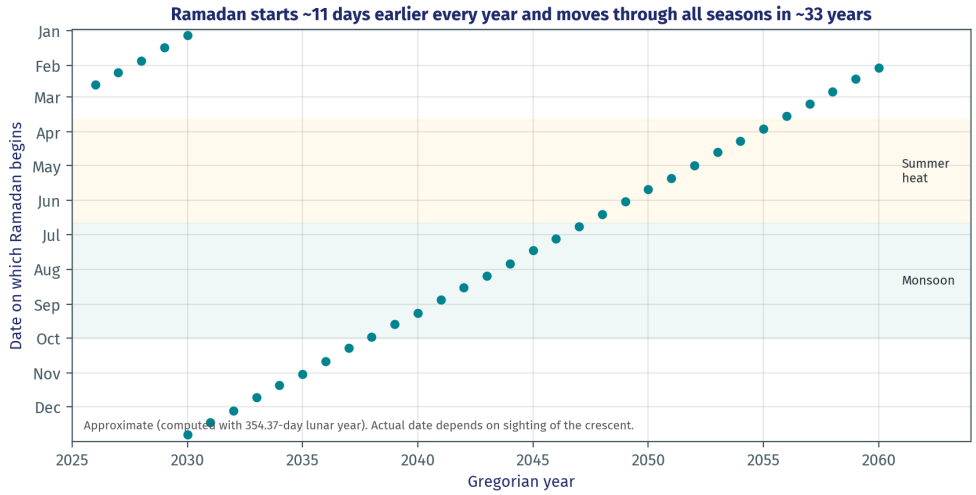
For calculations without sighting, a rule-based calendar uses a **30-year cycle** of 19 years of 354 days and 11 leap years of 355 days:

$$30 \text{ years} = 19 \times 354 + 11 \times 355 = 10{,}631 \text{ days}, \qquad \text{mean month} = \frac{10{,}631}{360} = 29.5306 \text{ days},$$

extremely close to the true synodic month (29.53059 days).

5. Ramadan through the seasons

$365.24 - 354.37 \approx 10.9$ days $\implies$ every Islamic date comes ~11 days earlier each Gregorian year.



Approximate start of Ramadan each year: it slides backwards through the calendar and returns after ~33 years.

Gregorian year	Ramadan begins (approx., ± 1 day with sighting)
2026	18–19 February
2027	~8 February
2028	~28 January
2029	~16 January
2030	~5 January **and** ~26 December — two Ramadans in one year!
2031	~15 December

A full cycle through the seasons takes $\dfrac{365.24}{10.88} \approx 33.6$ years: someone who fasts for 33 years will have fasted in every season.

6. Converting years

Since 33 Hijri years $\approx$ 32 Gregorian years:

$$\boxed{\text{AH} \approx (\text{CE} - 622) \times \frac{33}{32}} \qquad \boxed{\text{CE} \approx \text{AH} \times \frac{32}{33} + 622}$$

Worked examples

1. Late 2026: $(2026 - 622) \times \frac{33}{32} = 1404 \times 1.03125 = 1447.9 \Rightarrow$ **1448 AH** (it began in June 2026 ✔).

2. When did 1400 AH begin? $1400 \times \frac{32}{33} + 622 = 1357.6 + 622 = 1979.6 \Rightarrow$ late **1979** (1 Muharram 1400 AH = 21 November 1979 ✔).

3. Someone has lived exactly 33 Gregorian years: $\dfrac{33 \times 365.24}{354.37} \approx$ **34.0** Hijri years.

7. Astronomy in the service of daily life

- **Prayer times.** The five daily prayers are set by the Sun's position: Fajr (dawn twilight), Dhuhr (just after noon), Asr (when a shadow reaches a set length), Maghrib (sunset) and Isha (end of twilight). Mosques employed timekeepers (*muwaqqit*).
- **Qibla.** The direction of Mecca from any city is found with spherical trigonometry.
- **Instruments.** The **astrolabe**, an analogue computer for the sky, was perfected by Islamic-era craftsmen; in the Mughal period Lahore became a famous centre of astrolabe making.

Star names — an Arabic legacy in our sky

Star	Arabic origin	Meaning
Aldebaran	*al-dabaran*	"the follower" (it follows the Pleiades)
Betelgeuse	*yad al-jawza'*	"hand of al-Jawza"' (a giant figure)
Rigel	*rijl*	"foot"
Altair	*al-ta'ir*	"the flying (eagle)"
Vega	*al-waqi'*	"the swooping (eagle)"
Deneb	*dhanab*	"tail"
Fomalhaut	*fam al-hut*	"mouth of the fish"
Algol	*ra's al-ghul*	"head of the demon" (a star that "blinks")

Islamic-era astronomers

Abd al-Rahman al-Sufi (Persia, 964 CE) wrote the beautifully illustrated *Book of Fixed Stars*, which contains the first recorded description of the **Andromeda Galaxy** — a "little cloud". In 1420s Samarkand, **Ulugh Beg** built a giant observatory and produced one of the most accurate star catalogues made before the telescope.

A related calendar: the Persian Solar Hijri (and Parsi Navroz)

Iran and Afghanistan use a **solar** Hijri calendar. It also counts from 622 CE, but its year begins exactly at the **March equinox** (*Nowruz*), decided by observation — one of the most accurate calendars in the world. In India the **Parsi** community celebrates Navroz: Jamshedi Navroz at the March equinox, and Parsi New Year (Shahenshahi calendar) around August.

Key points to remember

- Hijri calendar: **purely lunar**, 12 months, **354/355 days**, **no leap months**.
- Epoch: the **Hijra, 622 CE**; 1448 AH began in June 2026.
- Months begin with the **sighting of the crescent (hilal)**; the day begins at **sunset**.



Left: Orion in a manuscript of al-Sufi's *Book of Fixed Stars* (Bibliothèque nationale de France, public domain). Right: the same figure placed on the real sky by Stellarium's "Arabic al-Sufi" sky culture (compiled by Khalid Al-Ajaji, CC BY-ND 4.0).

Key points to remember *(continued)*

- Dates move ~11 days earlier each year; Ramadan passes through all seasons in ~33 years.
- AH $\approx$ (CE $-$ 622) $\times \frac{33}{32}$.
- Many star names (Aldebaran, Altair, Vega, Deneb, Rigel, Betelgeuse) are **Arabic**.

Check yourself

1. Why does Eid al-Fitr come about 11 days earlier every year?
2. Why can Eid fall on different days in Kerala and in Saudi Arabia?
3. Roughly in which Gregorian year will 1500 AH begin?
4. What does "Altair" mean?

Answers: 1. The lunar year is ~10.9 days shorter than the solar year. 2. The crescent may be visible from one place and not the other (position, weather, differing criteria). 3. $1500 \times \frac{32}{33} + 622 \approx 2076$. 4. "The flying (eagle)".

Further reading (free / public)

- OpenStax, *Astronomy 2e*, §4.4 "The Calendar" — `https://openstax.org/books/astronomy-2e/pages/4-4-the-calendar`
- HM Nautical Almanac Office, crescent-visibility predictions — `https://astro.ukho.gov.uk/moonwatch/`
- E. G. Richards, *Mapping Time: The Calendar and its History* (Oxford University Press).